# Motorcycle Helmet Detection Using YOLOv8

Final Project Report

This project develops an object-detection model to identify motorcycle riders wearing helmets and riders without helmets.

## 1. Project Objectives

The objective is to train and evaluate a YOLOv8 object-detection model using two classes: With Helmet and Without Helmet.

## 2. Dataset Description

| Dataset split | Images |
|---|---|
| Training | 140 |
| Validation | 40 |
| Testing | 20 |
| Total | 200 |
| Classes | With Helmet; Without Helmet |
| Valid annotated boxes | 857 |

The dataset was organized into training, validation and testing splits using a 70/20/10 distribution. Dataset structure and label format were programmatically checked.

## 3. Model and Training Configuration

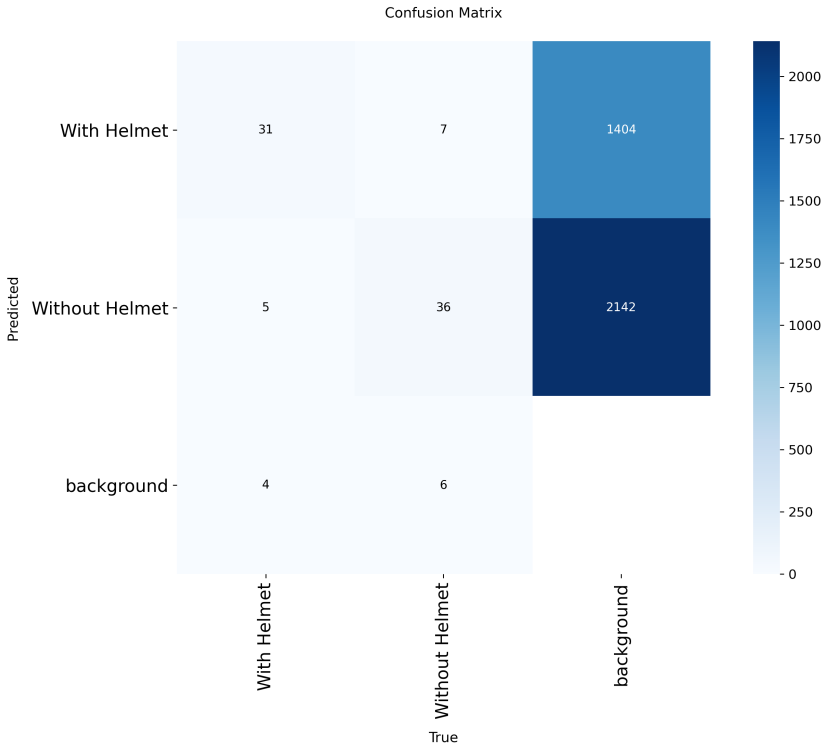
Model: YOLOv8n (pretrained weights)
Image size: 640 × 640
Epochs: 30
Batch size: 8
GPU: NVIDIA Tesla T4
Training time: approximately 4.27 minutes
Ultralytics version: 8.4.211

## 4. Test Set Evaluation

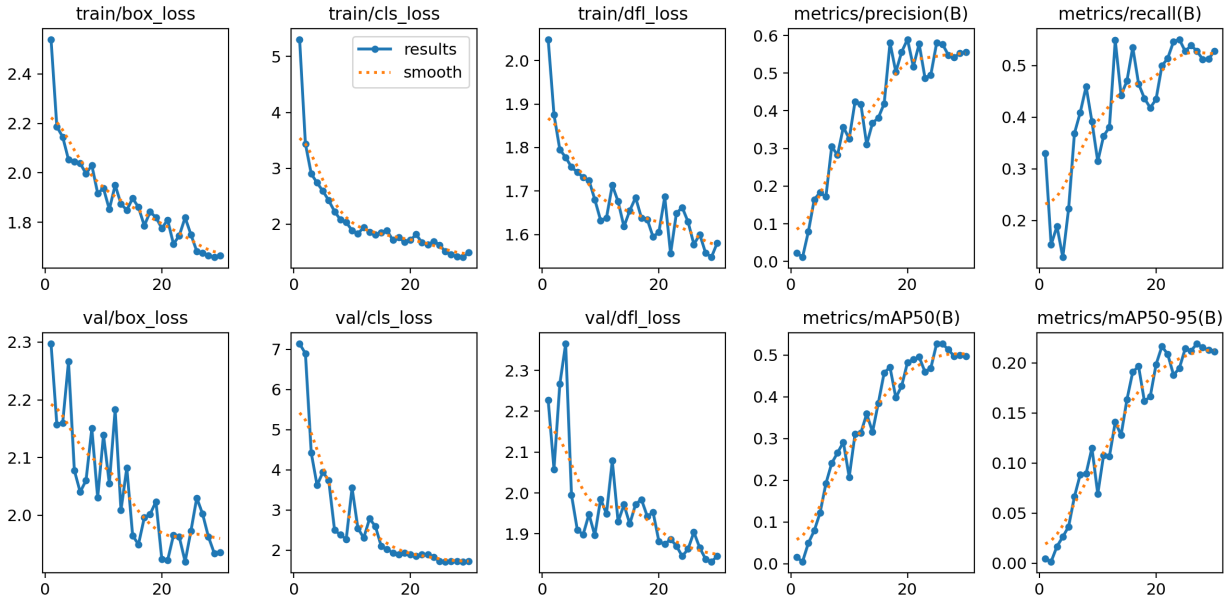
| Metric | Test result |
|---|---|
| Precision | 69.8% |
| Recall | 59.6% |
| mAP@0.5 | 65.0% |
| mAP@0.5:0.95 | 27.7% |

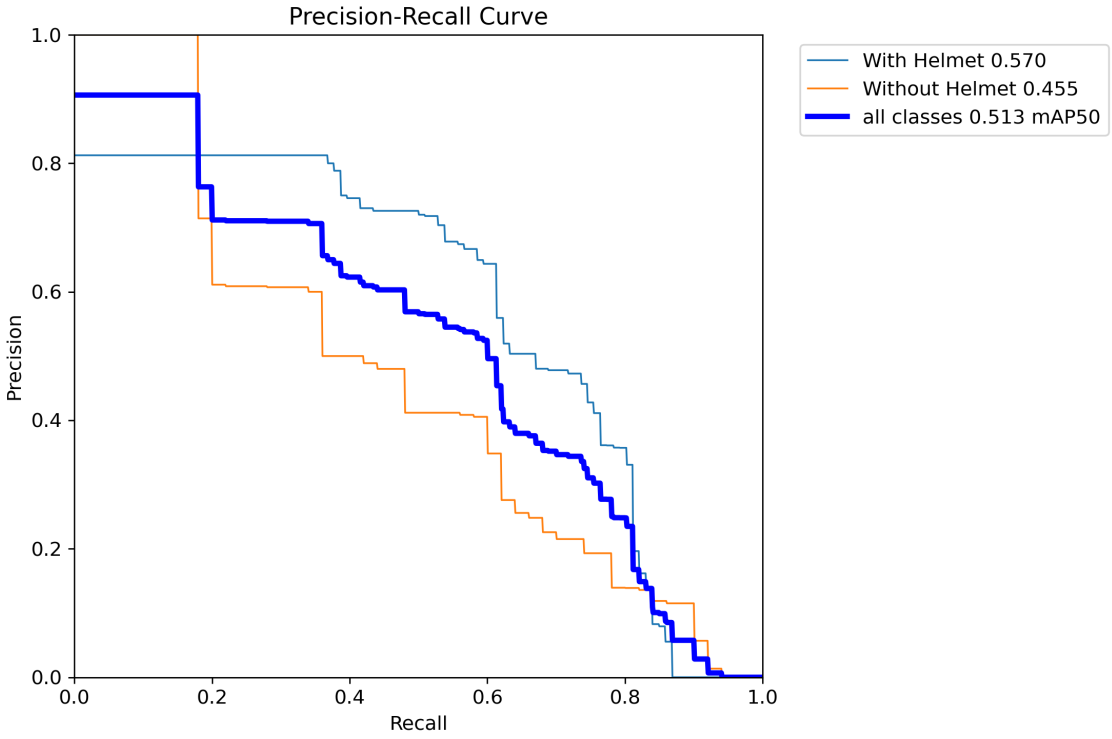
Class-wise test results:
With Helmet: Precision 57.1%, Recall 60.0%, mAP@0.5 62.9%.
Without Helmet: Precision 82.5%, Recall 59.2%, mAP@0.5 67.2%.

## 5. Confusion Matrix




## 6. Training Curves






## 7. Prediction Results

The trained model was applied to all 20 images in the test split. Annotated prediction images containing predicted classes, bounding boxes and confidence scores were saved in outputs/predictions.

manual__BikesHelmets115_png_jpg.rf.e2d44812a061abe2376fea5b3f9dcdc6.jpg




manual__BikesHelmets118_png_jpg.rf.276fb4c4dc04d2bb205f3790db6968ff.jpg




manual__BikesHelmets128_png_jpg.rf.689974532482e03a0a3b4362a02c8d3b.jpg




manual__BikesHelmets147_png_jpg.rf.325efbcc48fe92642606c8249b03f457.jpg




## 8. Automated Error Analysis

| Error category | Count |
|---|---|
| Correct detection | 32 |
| False Negative | 13 |
| False Positive | 12 |
| Wrong class | 7 |
| Total records | 64 |

These counts were produced by an automated comparison of predictions with ground-truth labels on the first 10 test images, using an IoU matching threshold of 0.5. They are detection records, not image counts. A separate manual visual review of 10 incorrect or weak predictions has not yet been completed.

## 9. Challenges and Limitations

The test results show that the model does not detect every annotated object. False positives, missed detections and wrong-class predictions were observed in the automated comparison. The dataset is relatively small, and the model's performance may be affected by image quality, object size, lighting, occlusion and annotation quality. These are possible factors and require further visual investigation.

## 10. Proposed Improvements

1. Add more diverse and accurately annotated training images.
2. Review bounding boxes and class labels to improve annotation quality.
3. Compare larger YOLO models and tune training settings, including epochs, batch size and confidence threshold.

## 11. Project Deliverables

The project folder contains the dataset, training notebook, trained best.pt model, training outputs, test evaluation outputs, 20 prediction images, automated error-analysis CSV, README.md and requirements.txt.

## 12. Conclusion

A YOLOv8n helmet-detection model was trained for 30 epochs and evaluated on a separate test split. The test mAP@0.5 was 65.0%. The project demonstrates an end-to-end object-detection workflow, while also identifying areas that need further improvement.

# Correction and Update: Manual Error Review

A visual review was performed on the error-heavy test image. Ten error records from this image were examined alongside the actual annotations and model predictions.

| Error Type | Reviewed Records |
|---|---|
| Wrong Class | 2 |
| False Positive | 4 |
| False Negative | 4 |
| Total | 10 |

Possible explanations include class confusion, unmatched detections, and missed objects. These are potential explanations; the exact causes were not independently established.

## Proposed Improvements

• Increase and balance the training dataset.

• Improve bounding-box and class-label annotation quality.

• Tune training and inference settings using validation data.

# Selected Error Review — 10 Records

Ten error records selected from five test images (BikesHelmets115, 128, 147, 215 and 237). Error categories originate from automated prediction-to-annotation matching and were considered alongside the visual observations discussed during review.

| Image | Actual | Prediction | Error type | Review status |
|---|---|---|---|---|
| BikesHelmets115 | Without Helmet | With Helmet | Wrong class | Selected for review |
| BikesHelmets115 | Without Helmet | With Helmet | Wrong class | Selected for review |
| BikesHelmets115 | No matched object | With Helmet | False Positive | Selected for review |
| BikesHelmets115 | No matched object | With Helmet | False Positive | Selected for review |
| BikesHelmets115 | No matched object | With Helmet | False Positive | Selected for review |
| BikesHelmets128 | Without Helmet | Not detected | False Negative | Selected for review |
| BikesHelmets147 | No matched object | With Helmet | False Positive | Selected for review |
| BikesHelmets147 | No matched object | With Helmet | False Positive | Selected for review |
| BikesHelmets215 | With Helmet | Without Helmet | Wrong class | Selected for review |
| BikesHelmets237 | With Helmet | Without Helmet | Wrong class | Selected for review |

## Review limitation

This is a selected review sample, not an independent visual audit of every prediction. Some automated records may not fully match the visual observations. Possible causes such as class confusion, missed objects, background detections, small objects and image blur should be treated as hypotheses unless individually confirmed.

## Recommended Model Improvements

• Increase training-data variety across camera angles, lighting and crowded scenes.

• Review and correct inconsistent class labels and bounding boxes.

• Experiment with suitable augmentation and tune inference confidence settings.